



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
EMS Academic Affairs Committee & Hon. Sophia Ragaza

“Multiplication and Division of Polynomials”
Practice Problems

1. Multiply: $3x(2x + 5)$
2. Multiply: $(x + 3)(x + 5)$
3. Multiply: $(2x^2 - 3x + 4)(x^2 + 2x - 5)$
4. Divide: $\frac{x^2 + 7x + 12}{x + 3}$
5. Divide: $\frac{2x^2 + 7x + 3}{2x + 1}$
6. Divide: $\frac{x^3 + 2x^2 - 5x - 6}{x + 3}$
7. Divide: $\frac{24x^5 - 18x^4 + 12x^3 - 6x^2}{6x^2}$
8. Multiply: $(2x - 3)(x^2 + 4x - 5)$
9. Multiply: $(3x^2 - 2x + 1)(x + 4)$
10. Divide: $\frac{3x^3 - 5x^2 - 10x + 8}{x - 2}$



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“Multiplication and Division of Polynomials”
Step-by-step Solution

1. Multiply: $3x(2x + 5)$

- **Rule:** Use the Distributive Property.

- **Step 1:** Multiply $3x$ by $2x$:

$$\text{➤ } 3x(2x) = 6x^2$$

- **Step 2:** Multiply $3x$ by 5 :

$$\text{➤ } 3x(5) = 15x$$

- **Step 3:** Combine:

$$\text{➤ } 6x^2 + 15x$$

Answer: $6x^2 + 15x$

2. Multiply: $(x + 3)(x + 5)$

- **Rule:** Multiply each term in the first polynomial by each term in the second polynomial.

- **Step 1:** $(x)(x) = x^2$

- **Step 2:** $(x)(5) = 5x$

- **Step 3:** $(3)(x) = 3x$

- **Step 4:** $(3)(5) = 15$

- **Step 5:** Combine like terms

$$\text{➤ } x^2 + 5x + 3x + 15$$

$$\text{➤ } x^2 + 8x + 15$$

Answer: $x^2 + 8x + 15$



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3. Multiply: $(2x^2 - 3x + 4)(x^2 + 2x - 5)$

- **Rule:** Use the Distributive Property. Multiply every term of the first polynomial by every term of the second polynomial.

- **Step 1:** Multiply $2x^2$:

$$\text{➤ } 2x^2(x^2 + 2x - 5)$$

$$\text{➤ } 2x^4 + 4x^3 - 10x^2$$

- **Step 2:** Multiply $-3x$

$$\text{➤ } -3x(x^2 + 2x - 5)$$

$$\text{➤ } -3x^3 - 6x^2 + 15x$$

- **Step 3:** Multiply 4:

$$\text{➤ } 4(x^2 + 2x - 5)$$

$$\text{➤ } 4x^2 + 8x - 20$$

- **Step 4:** Combine all terms:

$$\text{➤ } 2x^4 + 4x^3 - 10x^2 - 3x^3 - 6x^2 + 15x + 4x^2 + 8x - 20$$

- **Step 5:** Combine like terms:

$$\text{➤ } 2x^4 + x^3 - 12x^2 + 23x - 20$$

$$\text{Answer: } 2x^4 + x^3 - 12x^2 + 23x - 20$$

4. Divide: $\frac{x^2 + 7x + 12}{x + 3}$

- **Rule:** Factor the numerator first.

- **Step 1:** Find two numbers whose product is 12 and sum is 7:

$$\text{➤ } 3 \text{ and } 4$$



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- **Step 2:** Factor:

$$\triangleright x^2 + 7x + 12 = (x + 3)(x + 4)$$

- **Step 3:** Substitute:

$$\triangleright \frac{(x+3)(x+4)}{(x+3)}$$

- **Step 4:** Cancel the common factor:

$$\triangleright \frac{\cancel{(x+3)}(x+4)}{\cancel{(x+3)}}$$

$$\triangleright x + 4$$

Answer: $x + 4$

5. Divide: $\frac{2x^2 + 7x + 3}{2x + 1}$

- **Rule:** Factor the numerator.

- **Step 1:** Factor:

$$\triangleright 2x^2 + 7x + 3$$

$$\triangleright (2x + 1)(x + 3)$$

- **Step 2:** Rewrite:

$$\triangleright \frac{(2x+1)(x+3)}{2x+1}$$

- **Step 3:** Cancel the common factor:

$$\triangleright \frac{\cancel{(2x+1)}(x+3)}{\cancel{2x+1}}$$

$$\triangleright x + 3$$

Answer: $x + 3$



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6. Divide: $\frac{x^3 + 2x^2 - 5x - 6}{x + 3}$

- **Rule:** Use Polynomial Long Division:

$$\begin{array}{r} \overline{x^2 - x - 2} \\ x+3 \) \ x^3 + 2x^2 - 5x - 6 \\ \underline{x^3 + 3x^2} \\ -x^2 - 5x \\ -3x \\ -2x - 6 \\ \underline{-2x - 6} \\ 0 \end{array}$$

Answer: $x^2 - x - 2$

7. Divide: $\frac{24x^5 - 18x^4 + 12x^3 - 6x^2}{6x^2}$

- Rule: Divide each term in the polynomial by the monomial $6x^2$.
- **Step 1:** $\frac{24x^5}{6x^2} = 4x^3$
- **Step 2:** $\frac{-18x^4}{6x^2} = -3x^2$
- **Step 3:** $\frac{12x^3}{6x^2} = 2x$



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- **Step 4:** $\frac{-6x^2}{6x^2} = -1$
- **Step 5:** Combine the results:

$$\triangleright 4x^3 - 3x^2 + 2x - 1$$

Answer: $4x^3 - 3x^2 + 2x - 1$

8. Multiply: $(2x - 3)(x^2 + 4x - 5)$

- **Step 1:** Multiply $2x$ by each term:

$$\triangleright 2x(x^2 + 4x - 5)$$

$$\triangleright 2x^3 + 8x^2 - 10x$$

- **Step 2:** Multiply -3 by each term:

$$\triangleright -3(x^2 + 4x - 5)$$

$$\triangleright -3x^2 - 12x + 15$$

- **Step 3:** Combine

$$\triangleright 2x^3 + 8x^2 - 10x - 3x^2 - 12x + 15$$

- **Step 4:** Combine like terms:

$$\triangleright 2x^3 + 5x^2 - 22x + 15$$

Answer: $2x^3 + 5x^2 - 22x + 15$

9. Multiply: $(3x^2 - 2x + 1)(x + 4)$

- **Rule:** Use the Distributive Property.

- **Step 1:** Multiply $3x^2$:

$$\triangleright 3x^2(x + 4) = 3x^3 + 12x^2$$



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- **Step 2:** Multiply $-2x$:

$$\rightarrow -2x(x + 4) = -2x^2 - 8x$$

- **Step 3:** Multiply 1:

$$\rightarrow 1(x + 4) = x + 4$$

- **Step 4:** Combine:

$$\rightarrow 3x^3 + 12x^2 - 2x^2 - 8x + x + 4$$

- **Step 5:** Combine like terms:

$$\rightarrow 3x^3 + 10x^2 - 7x + 4$$

$$\text{Answer: } 3x^3 + 10x^2 - 7x + 4$$

10. Divide: $\frac{3x^3 - 5x^2 - 10x + 8}{x - 2}$

- **Rule:** Use Polynomial Long Division.

$$\begin{array}{r} \underline{3x^2 + x - 8} \\ x - 2 \end{array}$$

$$x - 2 \) \ 3x^3 - 5x^2 - 10x + 8$$

$$\underline{3x^3 - 6x^2}$$

$$x^2 - 10x$$

$$\underline{x^2 - 2x}$$

$$-8x + 8$$

$$\underline{-8x + 16}$$

$$-8$$

$$\text{Answer: } 3x^2 + x - 8 - \frac{8}{x - 2}$$